CIBC HACKATHON | NEXT BEST ACTION USECASE

Team : TrustIssues

Collections that are fair, explainable and effective

Know who the customer really is. Contact the right person, the right way, with clear reasoning an agent can read out loud.

1.01M
people, resolved from 3.49M source records

0.968
promise-break AUC, vs 0.712 for the bank's model

0
protected attributes used in any decision

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86,059 open cases, and data that the agents can't trust

56.7%

of promises to pay are broken today

28.5% → **35.3%**

Bank's model | Ours

caught in the riskiest 20% (35.3% is the maximum possible at this broken rate)

Precision in the riskiest 20%: 80.7% (bank) vs 99.9% (ours)

WHAT IT COSTS IN THE OLD SYSTEM

01 **Recovery lost**

Agent time goes to customers who would have paid anyway.

02 **Customer harm**

People in financial difficulty get a payment reminder instead of help.

03 **Compliance exposure**

Most customers appear as 3-4 records across 5 systems, so contact rules hit fragments, not the person.

Four layers, one pipeline, a person in the loop

01 **Data Product Factory**	Turns messy raw files into one trusted golden customer view, with a match confidence on every link.
02 **Insight & NLP (Ask)**	Anyone can ask a plain-English question and see the SQL or source behind every answer. (Ensures role based access)
03 **Feature Store**	One definition of each signal, from numbers, notes and calls, used identically in training and live scoring.
04 **Models & Decisioning**	Every day it shows the recommended action per case, with a plain-English reason and a human check.

Layer 1: one person, one history

3.49M

source records from five systems

→

1.01M

people, each link carrying a confidence

+1,443

open cases whose CRM record doesn't exist can now be worked (before: 6)

Matching ladder

Exact rules first, then Splink, then an LLM second opinion, then a human. When unsure, records stay separate.

Confidence per link

1.00 for CIF and national ID, 0.99 for name + DOB + phone, and a match probability for Splink.

Gated publishing

73 issues documented across 31 tables, nothing deleted. Gold updates only when every blocking rule passes.

1,011,960 people found vs 1,000,000 stated in the release; we report the gap and send grey-zone pairs to human review.

Beyond the brief, on every layer

Identity: four-stage matching ladder with a human review queue

Ask: certified metrics beside validated, read-only SQL

Evidence-or-drop text features: no exact quote in the source, no feature

Rules before models: a model can never pick a forbidden action

Decisions: SHAP reasons, runner-up, blocked actions, logged overrides

OUR SIGNATURE IDEA

Tagged-record matching

Both records become one tagged line, so the reader knows which system each field came from.

```
[SRC] card [NAME] R MITCHELL [DOB] 1988-03-14
[PHONE] 519555xxx42 [SEP] [SRC] crm [NAME]
RYAN J MITCHELL …
```

It copes with initials, nicknames, day/month swaps and masked digits, and never links alone: any disagreement goes to a human.

Inspired by MAMMAL (npj Drug Discovery) and Ditto (VLDB 2021): we borrow the idea, not their models

Layer 2: ask in plain English, via SQL or RAG

What share of outbound calls in September 2026 were outside permitted calling hours, by customer province?

Certified metric As of 2026-09-28

Province	Out of hours pct	Calls
AB	0	20,991
BC	0	25,977
MB	0	5,107

26/26
unseen questions correct

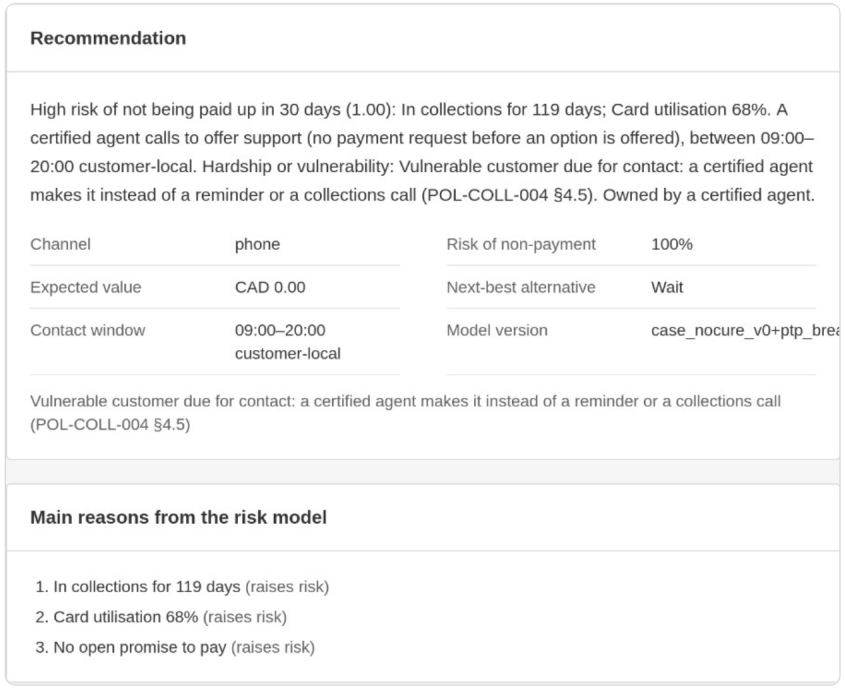
50/50
benchmark questions submitted

15/16
policy questions, right section found

NL → SQL + two RAG types
SQL shown on every answer; BM25 and hybrid + rerank retrieval; refuses out-of-scope.

Layers 3–4: decisions an agent can read aloud (NBA Use Case)

Features → **Rules first** → **Risk models** → **Action value** → **Human review**



Recommendation

High risk of not being paid up in 30 days (1.00): In collections for 119 days; Card utilisation 68%. A certified agent calls to offer support (no payment request before an option is offered), between 09:00–20:00 customer-local. Hardship or vulnerability: Vulnerable customer due for contact: a certified agent makes it instead of a reminder or a collections call (POL-COLL-004 §4.5). Owned by a certified agent.

Channel	phone	Risk of non-payment	100%
Expected value	CAD 0.00	Next-best alternative	Wait
Contact window	09:00–20:00 customer-local	Model version	case_nocure_v0+ptp_brea

Vulnerable customer due for contact: a certified agent makes it instead of a reminder or a collections call (POL-COLL-004 §4.5)

Main reasons from the risk model

1. In collections for 119 days (raises risk)
2. Card utilisation 68% (raises risk)
3. No open promise to pay (raises risk)

Measure	Bank	Ours
Case not cured in 30 days, AUC	0.725	0.839
Promise break, AUC*	0.712	0.968
Precision in the riskiest 20%	80.7%	99.9%
Broken promises caught in riskiest 20%	28.5%	35.3%†

AUC: how well a model ranks risky cases above safe ones (0.5 = coin toss, 1.0 = perfect).

† The maximum possible at this broken rate (20% ÷ 56.7%).

* Most of the promise-model lead comes from the promised share of the overdue amount; without those features AUC is 0.674. Checked: not back-filled from the outcome.

Responsible by design, with people in control

Fairness: tested by group, flags published

Gender, age, language and similar attributes are never model inputs. We join them back only to test whether outcomes still differ by group.

A flag means “worth a human look”, not a legal finding: calibration off by over 0.05, group AUC over 0.05 below overall, selection under 80% of the best group, or features that predict the attribute (AUC 0.6+).

Model	Flags	Where
Promise-break (ours)	1	Preferred language
Case-risk (ours)	5	Age, newcomer, language
Case-risk (bank's)	2	Age, marital status

Groups genuinely differ in how often they break or cure, so calibration within each group is the better test.

Hardship means more help

No reminders or collection calls; a certified agent offers an eligible programme.

Humans stay in control

Overrides logged with a reason code, reviewed weekly; team leads approve sensitive actions.

Secure by default

Role-based access, MFA, rate limits, audit logs; text masked before it leaves the machine.

Tech stack, each choice with a reason

ENGINE

DuckDB + Parquet

1M customers and 12M-row tables fit on one machine: fast SQL, no cluster.

IDENTITY

SQL rules + Splink

Cheapest, surest step first; an LLM and a human handle the grey zone.

FEATURE STORE

Feast + SQL files

The same SQL feeds training and live scoring; no real-time store needed.

MODELS

LightGBM + SHAP

Strong on tabular data; SHAP turns each score into plain-English reasons.

ASK

Templates + sqlglot

Certified metrics stay consistent; generated SQL is validated read-only.

APP

FastAPI + React

Approve and override single rows, role views and streaming answers.

How we will measure impact, and what comes next

WHAT WE MEASURE

Promise-kept rate

Also: 30-day cure rate, early roll rate, contacts per dollar recovered.

Run as a challenger with a permanent 5% hold-out; complaints, opt-outs and hardship offers as guardrails.

NEXT: THE REST OF THE BRIEF

Next	**Post-call summary and QA automation**, built on the call transcripts we already read.
Then	**Agent assist and routing:** live in-call prompts, and cases matched to the best agent.
Later	**Voice recordings** as features, plus online feature serving for real-time scoring.

> **A collections team that knows who the customer is, helps those in difficulty first, and can explain every decision.**

Repo: https://github.com/Checha18/cibc-trustissues
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